



Hands-On Activity: Clean data with spreadsheet functions

Activity Overview

By now, you've been introduced to some useful techniques for cleaning spreadsheet data, such as sorting and filtering. In this activity, you'll continue to develop your data-cleaning skills by using spreadsheet functions.

Review the following scenario. Then complete the step-by-step instructions.

Imagine you are working for a marketing agency based in San Francisco. The marketing agency wants to contact local boba tea shops to inquire about a potential collaboration for a new marketing campaign. The agency plans to visit the top-rated shops within a 10-mile radius of the center of their target area. To assist with planning, the agency asks your team to review external data related to ratings and locations of boba tea shops in San Francisco. One of your teammates has created a spreadsheet from an online source. However, the data is not in the greatest shape.

Your assignment is to identify the dirty elements in the dataset and clean them up.

By the time you complete this activity, you will be able to identify dirty elements in a dataset, remove duplicate data, and use the **COUNTIF** and **SPLIT** functions to help clean data.

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

To get started, access the spreadsheet that contains the data. Click the link and make a copy of the [spreadsheet](#).

Or, if you don't have a Google account, you may download the dataset directly from the attachment below:

The dataset includes the following column headers:

Column Header	Description
id	a unique identifier for each boba shop
name	name of boba shop
rating	Yelp rating (0 to 5 stars)
address	street address
city	city
lat-long	latitude and longitude

Your job is to present data that is readable, accurate, and visually appealing. Cleaning your data helps you achieve this goal. The first step is to identify the dirty elements in your data.

1. Rename your spreadsheet. Click **Untitled Spreadsheet** and enter a new name. You can use the name **sf_boba_tea_shop_data** or a similar name that describes the data your spreadsheet contains.
2. If you want to get a better view of your data, you can make the columns wider by dragging the right boundary of the column heading. This may apply to the **name** (B), **address** (D), and **lat-long** (F) columns.
3. Now, review your data and consider any problems you may need to address. The following are examples of errors that you can quickly identify and fix. This is not a comprehensive list of every potential problem, but is a great starting point for data cleaning.
 - First, there is at least one duplicate line (rows 20 and 21) in your dataset.

20	17	mandro-teahouse-newark-3	4	34956 Newark Blvd	Newark	37.5515049151237-122.050272187505
21	17	mandro-teahouse-newark-3	4	34956 Newark Blvd	Newark	37.5515049151237-122.050272187505

- Second, all Yelp ratings should fall between 0 and 5. However, at least one rating (in cell C8) falls outside of that range.

	A	B	C	D	E	F
1	id	name	rating	address	city	lat-long
2	0	99-tea-house-fremont-2	4.5	3623 Thornton Ave	Fremont	37.56295-122.010039999999
3	1	one-tea-fremont-2	4.5	46809 Warm Springs Blvd	Fremont	37.4890666928572-121.929413750767
4	2	royaltea-usa-fremont	4	38509 Fremont Blvd	Fremont	37.5513151288032-121.993849799037
5	3	teco-tea-and-coffee-bar-fremont	4.5	39030 Paseo Padre Pkwy	Fremont	37.5536945-121.981043
6	4	t-lab-fremont-3	4	34133 Fremont Blvd	Fremont	37.576149-122.0437049
7	5	q-tea-monster-newark	4	39181 Cedar Blvd	Newark	37.5229604101756-122.005785632481
8	6	gong-cha-fremont	6.7	46827 Warm Springs Blvd	Fremont	37.4885682635695-121.929191268869

- Finally, the data for latitude and longitude is contained in a single column (F). In order for someone to be able to use this data for analysis, the two values should be in separate columns.

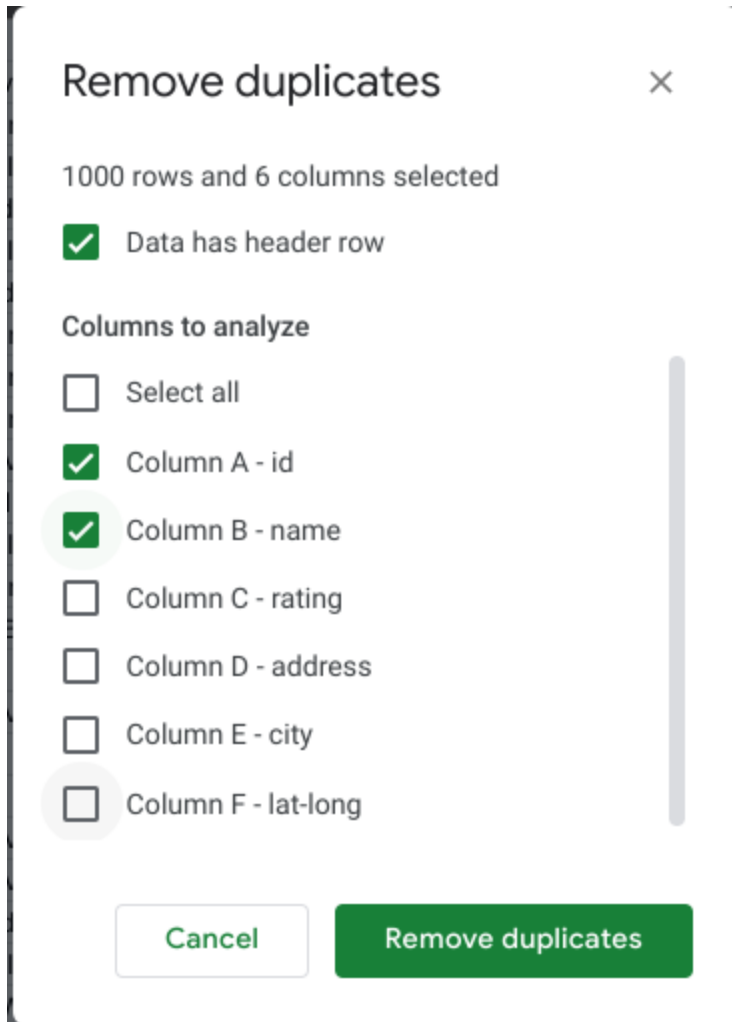
F	G	H
lat-long		
37.56295-122.010039999999		
37.4890666928572-121.929413750767		
37.5513151288032-121.993849799037		
37.5536945-121.981043		
37.576149-122.0437049		
37.5229604101756-122.005785632481		
37.4885682635695-121.929191268869		
37.4885682635695-121.929191268869		
37.4884429093476-121.930383669657		

Now you know what issues to focus your attention on during the cleaning process.

Your goal is to fix these errors and help create a clean dataset for analysis. You can address each issue in turn.

The first step is to eliminate any duplicate entries from your dataset. As a best practice, duplicates should be removed even if they are not readily apparent.

1. To start, select columns A through F.
2. Then, in the menu bar, choose **Data**, then **Data Cleanup**, and select **Remove duplicates**.
3. In the pop-up window, click **Data has header row**. You want to remove duplicate boba shop id's and boba shop names. In the **Columns to analyze** section, make sure the relevant columns (**id**, **name**) are selected.



4. Once everything has been selected, click **Remove duplicates**.

5. If done correctly, 3 duplicate rows will be found and removed and 604 rows will remain.

Next, clean up any data that does not make sense. Yelp ratings should be less than 5 and greater than 0. Now, you will determine how many entries are inaccurate and correct them. You can use the **COUNTIF** function to perform this task.

1. The **COUNTIF** function quickly counts how many items in a range of cells meet a given criterion. In cell I2, enter **=COUNTIF(C:C, ">5")**. The first entry (**C:C**) refers to the range where you are counting the data. In this case, the range is the entire **rating** column (C), which contains the Yelp ratings. The second entry refers to the criterion (**>5**), and tells the function to count all the values greater than 5.
2. Press **Enter**. You'll notice that the function returns a value of 9. This tells you that your dataset contains 9 entries that have a rating greater than 5.

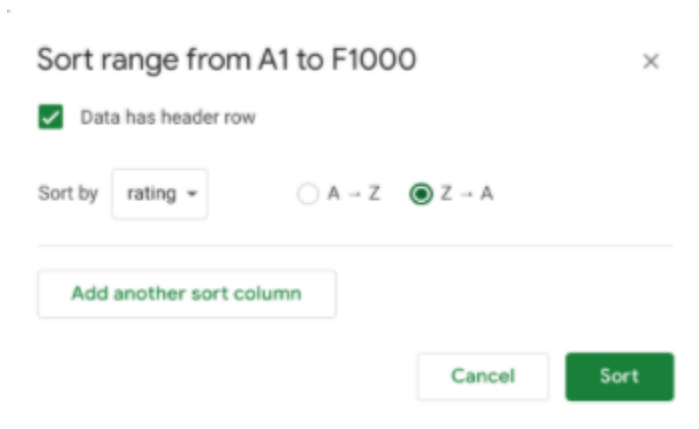
I2	+ fX	=COUNTIF(C:C,">5")						
	C	D	E	F	G	H	I	
1	rating	address	city	lat-long				
2	4.5	3623 Thornton Ave	Fremont	37.56295-122.010039999999			9	
3	4.5	46809 Warm Springs Blvd	Fremont	37.4890666928572-121.929413750767				
4	4	38509 Fremont Blvd	Fremont	37.5513151288032-121.993849799037				

It's your job to decide what to do with incorrect values or to ask the dataset owner for advice if you're unsure. In this case, one effective approach would be to search on Yelp for the actual ratings. For this activity, you can just replace the incorrect ratings with the number 5. An efficient way to replace the ratings is to sort the data numerically from largest to smallest rating.

3. Select columns A through F.

4. Then, from the menu bar, choose **Data**, then **Sort range**, and select **Advanced range sorting options**.

5. In the pop-up window, check the box next to **Data has header row**. Sort by **rating** from **Z → A**. This way, the highest ratings will be listed first.



6. Click **Sort**. Check out your spreadsheet. At the start of the **rating** column, you should now find the 9 rows that have incorrect values (rating > 5).

	A	B	C
1	id	name	rating
2	243	che-lo-union-city-2	9.2
3	88	super-cue-cafe-san-francisco-2	8.9
4	133	t4-san-leandro	7.4
5	6	gong-cha-fremont	6.7
6	271	happy-lemon-sunnyvale-2	6.2
7	218	ohana-hawaiian-bbq-of-pleasanton-pleasanton	5.7
8	65	infinitea-san-francisco	5.6
9	160	amor-cafe-and-tea-san-jose	5.4
10	23	boba-queen-fremont	5.2

7. Next, select the range of cells **C2:C10**. Press **delete** to delete the values that are greater than 5.

8. Replace all the values with the number **5**. In cell C2, enter **5**. Then, drag the fill handle down to cell C10 to fill the remaining cells with **5**.

9. After replacing the incorrect ratings with the number 5, you may notice that the new value in cell I2 is 0. The output of the **COUNTIF** function now reflects the changes in your dataset. This confirms that the **rating** column no longer contains any values greater than 5.

10. Finally, delete the formula from cell I2 since you don't need this information anymore.

Next, clean up the latitude and longitude data by placing each value in a separate column. You can use the **SPLIT** function to accomplish this task.

1. The **SPLIT** function divides text around a specified character or string, and puts each fragment of text into a separate cell in the row. The **SPLIT** function will split the single **lat-long** column into two separate columns, one for latitude and the other for longitude. In cell G2, enter **=SPLIT(F2, "-")**. The first entry (**F2**) refers to the cell where the text is located. The second entry ("**-**") refers to the fact that you are dividing the text based on the minus sign.

F	G	H
lat-long		
37.5895628278523-122.022492714298	=SPLIT(F2, "-")	
37.7242954229777-122.457044541931		

2. Press **Enter**. The result shows each fragment of text in a different cell.

F	G	H
lat-long		
37.5895628278523-122.022492714298	37.58956283	122.0224927
37.7242954229777-122.457044541931		

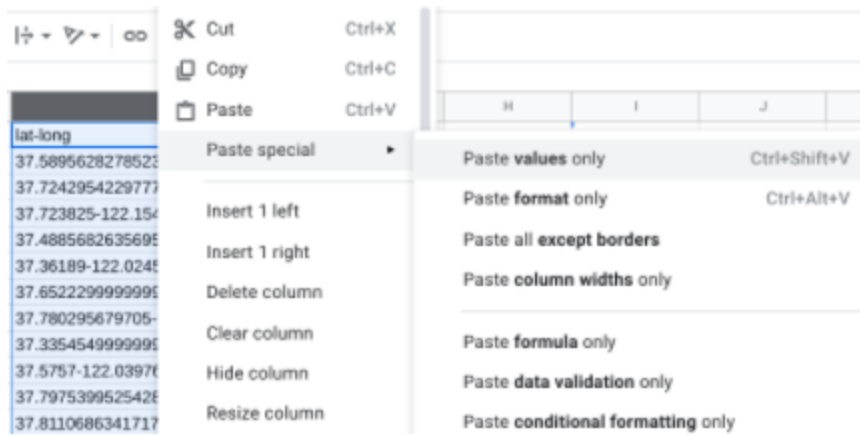
3. Select cell G2 again. In cell G2, double-click on the fill handle to split all the remaining **lat-long** entries.

4. Now add column headers to the two new columns (G and H). In cell G1, enter **lat**. In cell H1, enter **long**.

5. Next, replace the original **lat-long** data in column F with the new split entries in columns G and H. Select columns G and H, right-click, and choose **Copy**.

F	G	H
lat-long	lat	
37.5895628278523-122.022492714298	37.58956283	
37.7242954229777-122.457044541931	37.72429542	
37.723825-122.154662999999	37.723825	
37.4885682635695-121.929191268869	37.48856826	
37.36189-122.024539999999	37.36189	
37.6522299999999-121.8786	37.65223	
37.780295679705-122.477084781597	37.78029568	
37.3354549999999-121.886596	37.335455	
37.5757-122.039769999999	37.5757	
37.7975399525428-122.406789958477	37.79753995	
37.8110686341717-122.24723573774	37.81106863	

6. Then, select Column F, right-click, and choose **Paste special** and **Paste values only**.



7. Now the new **lat** column is column F, and the new **long** column is column G. Adjust the width of the **lat** column (F) to fit the data by dragging the right boundary of the column heading.

F	G
lat	long
37.58956283	122.0224927
37.72429542	122.4570445
37.723825	122.154663
37.48856826	121.9291913
37.36189	122.02454
37.65223	121.8786
37.78029568	122.4770848
37.335455	121.886596
37.5757	122.03977

8. Next, select column H, right-click, and choose **Delete column**.

9. Finally, the longitude values should be negative so that they are accurate coordinates for mapping. To make the values in the **long** column negative, multiply them by **-1**. In cell H2, enter **=G2*-1**. The asterisk is the operator for multiplication. Press **Enter**.

10. Still in cell H2, double-click on the fill handle to fill in the rest of the values.

11. Next, add a column header. In cell H1, enter: **long**.

12. Now, replace the longitude data in column G with the new data in column H. Select column H, right-click, and choose **Copy**.

13. Select Column G, right-click, and choose **Paste special** and **Paste values only**.

14. Then, select column H, right-click, and choose **Delete column**.

Columns F and G should look like this:

F	G
lat	long
37.58956283	-122.0224927
37.72429542	-122.4570445
37.723825	-122.154663
37.48856826	-121.9291913
37.36189	-122.02454
37.65223	-121.8786
37.78029568	-122.4770848
37.335455	-121.886596
37.5757	-122.03977

Now your data is cleaner, clearer, and easier to use.

Be sure to save a copy of the spreadsheet template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in the future.

1.

Question 1

Reflection

Which of the following functions divides text around a specified character or string and puts each fragment of text into a separate cell in the row?

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The **CONCATENATE** function

The **TRIM** function

The **SPLIT** function

The **COUNTIF** function

Correct

The **SPLIT** function divides text around a specified character or string, and puts each fragment of text into a separate cell in the row. Spreadsheet functions are useful tools for data cleaning, and knowing how to use functions effectively is a key part of every data analyst's skill set.

2.

Question 2

In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- As a data analyst, why is it important for the data you present to be readable, accurate, and visually appealing?
- How can spreadsheet functions help you clean data more efficiently and effectively?

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My Answer:

Without readable, accurate, and visually appealing data, as an analyst, it would not be possible to do analysis. Without this, we can't get any insight, and the total analysis step becomes worthless and erroneous. Yeah, in a spreadsheet we can do the cleaning process in a very easy way, and we have so many options and formulas to do this, so the cleaning process is smoother and becomes effective. And also we get an efficient result.

Feedback

Congratulations on completing this hands-on activity! In this activity, you used spreadsheet functions to clean “dirty” data in a spreadsheet.

Cleaning data is an important part of the data analysis process. If data analysis is based on bad or “dirty” data, it may be biased, erroneous, and uninformed. Knowing how to effectively use spreadsheet functions to work with data is an essential skill for every data analyst. In upcoming activities, you will continue to learn more about spreadsheet functions and how they can help you analyze your data.